RAUL ABABEI

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Education

University of Bucharest — Bucuresti, Romania
Bachelor's Degree in Computer Science - current GPA: 9.58/10.0 — *Sep 2025 – Present*

"Grigore Moisil" National College of Computer Science — Brasov, Romania
Highschool Diploma, Mathematics-Informatics; Final grade: 9.75 — *Sep 2021 – Jun 2025*

Experience

Hackathon Competitor — International / National
Selected AI, Machine Learning & Systems Competitions — *2025 – 2026*

- **CISPA European Cybersecurity & AI Championship (International):** 24-hour pan-European hackathon at KTH Royal Institute of Technology (Stockholm, Sweden); engineered solutions for model inversion, ML misclassification analysis, and label anomaly detection.
- **ITFest Adobe 2026 (2nd Place):** Engineered an automated competitive intelligence platform and multi-model AI pipeline to analyze competitor ads and synthesize tailored campaigns.
- **ElevenLabs AI Tinkerers (4th Place):** Co-developed *.flux*, a multimodal dashboard using AI agents to synthesize repository PRs and transcripts.

Mechanical Team Leader — Brasov, Romania
Kronbot - FIRST Tech Challenge Romania — *Sep 2022 – Jun 2024*

- Directed CAD modeling, rapid prototyping, and live match repairs for competitive robots.
- Integrated hardware with embedded software, securing qualification for the 2023 FTC National Championship.

Product Development (Intern) - "Fit4Future" program — Brasov, Romania
Preh Romania — *Mar 2024*

Projects

Rippletrace | *Next.js, FastAPI, Python, Neo4j, LangChain, Sec-api* | GitHub — Mar 2026 – May 2026

- Engineered a full-stack platform (Next.js, FastAPI) to model and visualize cascading supply chain and financial disruptions.
- Built a 3-phase automated pipeline streaming market data into a Neo4j knowledge graph, orchestrating GraphRAG with Llama 3 for real-time risk simulation.

.flux | *AI Agents, Multimodal Voice Assistant* | GitHub — Dec 2025

- Secured **4th place** at the ElevenLabs "AI Tinkerers" 4-hours Hackathon by co-developing an intelligent dashboard that unifies scattered engineering context.
- Orchestrated specialized AI agents to synthesize data from GitHub repositories, PRs, and meeting transcripts to create a semantic map of codebase evolution.

Quest | *Raspberry Pi, Arduino MEGA, YOLOv8n, PCB* | GitHub — Feb 2024 – Aug 2024

- Designed a hybrid search-and-rescue robot capable of morphing between a four-wheeled rover and a quadcopter drone with a custom 3D-printed PETG-CF chassis.
- Implemented an autonomous object detection system using YOLOv8n on an Oak-D Pro camera to identify and track victims in real-time.

AdReverse | *Next.js, FastAPI, LangChain, Groq, Apify, Supabase* | GitHub — Nov 2026

- Secured **2nd Place** at ITFest Adobe 2026 by engineering a platform that reverse-engineers competitor ads via an automated 4-stage pipeline and brand-tailored synthesis.
- Automated asset scraping with Apify and Tavily, orchestrating multi-LLM workflows via LangChain and Groq (with Gemini Flash fallback), Supabase, and Clerk.

Technical Skills

Languages: C++/C, Python, JavaScript, C#, PHP, HTML, CSS, PL/SQL, Assembly, Haskell
Frameworks/Libraries: React, Flask, FastAPI, ASP.NET, Next.JS, Supabase, Convex
Tools/Methodologies: Git, PostgreSQL, Linux, Neo4j, LangChain, PyTorch, Eigen, Docker
Spoken Languages: Romanian (native), English (C1)

Honors & Certifications

- **InfoEducație National Olympiad:** 3rd Place (2024)
- **Robochallenge International:** Honorable Mention, Freestyle Showcase (2024)
- **CS50's Introduction to Programming with Python:** HarvardX (2025)
- **AI Fundamentals: Foundations for Understanding AI:** IBM (2026)